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Storage of Individual UL2 and UL2.2 Aftertreatment Diesel Exhaust Fluid (DEF) System Components

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General Information

This document is intended for UL2 and UL2.2 aftertreatment diesel exhaust fluid (DEF) systems and will outline how to store individual components that are being stored with the intent of being reused after storage. Storage time is **not** to exceed two years. Because of the number of varying applications, this document has been written to be generic. **Not** all instructions within this document will represent the application being serviced.

Recommended Cummins® Service Tools

- INSITE™ electronic service tool

Additional Service Items

- Clean bucket, 1 gallon or more
- Double distilled water
- Aftertreatment clean care kit, Part Number 5299193
- Diesel exhaust fluid (DEF) clean care kit, Part Number 5394236
- Aftertreatment DEF dosing unit filter

Preparing for Storage of Individual Components

⚠ CAUTION ⚠

Aftertreatment DEF system is extremely sensitive to dirt and debris. All connections and ports must be cleaned and covered. Even a very small amount of debris can result in DEF system malfunction.

⚠ WARNING ⚠

DEF contains urea. Do not get the substance in your eyes. In case of contact, immediately flush eyes with large amounts of water for a minimum of 15 minutes. Do not swallow. In the event the DEF is ingested, contact a physician immediately. Reference the Materials Safety Data Sheet (MSDS) for additional information.

⚠ WARNING ⚠

The DEF line connecting the aftertreatment DEF dosing unit to the aftertreatment diesel exhaust fluid dosing valve is under low pressure and should not be disconnected while the engine is running or before the system has completed the purge process after engine shutdown. Disconnecting the DEF line while under low pressure could cause DEF to spray.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Note : Long term storage in a vehicle (in excess of 6 months) is **not** recommended. If long term storage is necessary, periodic testing of the DEF is recommended to verify the concentration does **not** fall out of specification. Reference Service Bulletin, Diesel Exhaust Fluid (DEF) Specifications for Cummins® Selective Catalytic Reduction (SCR) Systems, Bulletin 4021566.

Note : The following instructions provided include optional steps for allowing aftertreatment DEF dosing system components to be stored on the application:

1. Drain DEF from DEF tank and aftertreatment DEF dosing system components (optional). Follow local procedures for proper disposal of DEF.
2. Clean inside of DEF tank (optional). See corresponding Original Equipment Manufacturer (OEM) service procedure.
3. Fill DEF tank with double distilled water (if double-distilled water is **not** available, use distilled water). Alternately, fill a clean bucket with double distilled water and place DEF dosing unit suction line into bucket and remove return line from DEF tank and allow to fill an empty bucket.
4. Remove DEF dosing valve from aftertreatment decomposition tube/selective catalytic reduction (SCR) catalyst, do **not** remove DEF lines or wiring harness from DEF dosing unit or DEF dosing valve at this time. See corresponding Service Manual. Reference Procedure 011-059 in Section 11 for removal of DEF dosing valve from decomposition tube/aftertreatment SCR catalyst.
5. Activate DEF dosing system using INSITE™ electronic service tool to perform Diesel Exhaust Fluid Dosing Unit Override Test three times. See corresponding Service Manual. Reference Procedure 011-063 in Section 11.
6. Use INSITE™ electronic service tool to perform Aftertreatment Diesel Exhaust Fluid System Leak Test. Run DEF dosing unit dry until no fluid comes out of DEF dosing valve return line. See corresponding Service Manual. Reference Procedure 011-080 in Section 11.
7. Remove DEF lines from DEF dosing valve. Cap DEF dosing valve inlet and outlet and plug aftertreatment DEF lines with plugs from aftertreatment clean care kit, Part Number 5299193, to reduce possibility of debris entering system (water can stay inside of DEF dosing valve). Cover electrical connectors with tape, Part Number 4919420. See corresponding Service Manual. Reference Procedure 011-059 in Section 11. Place DEF dosing valve in cardboard or plastic box. Store in a clean and dry environment protected from weather. Do **not** store in direct sunlight (optional). Alternately, install DEF dosing valve to aftertreatment

decomposition tube/ SCR catalyst. Reference Procedure 011-059 in Section 11. Install all DEF lines and harness connectors.

8. Remove DEF suction line and pressure line from DEF dosing unit.
9. Remove main filter from DEF dosing unit. See corresponding Service Manual. Reference Procedure 011-060 in Section 11.
10. Use INSITE™ electronic service tool to perform Aftertreatment Diesel Exhaust Fluid System Leak Test while applying 94 psi [650 kPa] compressed air at DEF dosing unit inlet and outlet connectors for 40 seconds each to dry DEF dosing unit.
11. Install new main DEF dosing unit filter. See corresponding Service Manual. Reference Procedure 011-060 in Section 11.
12. Cover aftertreatment DEF connectors and exposed connections on aftertreatment DEF lines. Use protective caps from aftertreatment DEF clean care kit, Part Number 5394236. Cover electrical connectors with tape, Part Number 4919420 (optional). Alternately, install all DEF lines and harness connectors.
13. Place DEF dosing unit in cardboard or plastic box and store in a clean and dry environment protected from weather (optional). Do **not** store in direct sunlight.

Removing from Individual Storage

1. Clean inside of aftertreatment DEF tank (optional). See equipment manufacturer service information. Alternately, test the concentration of DEF, use Cummins® DEF refractometer, Part Number 4919554. Follow the instructions provided with the service tool. Reference Service Bulletin, Diesel Exhaust Fluid (DEF) Specifications for Cummins® Selective Catalytic Reduction (SCR) Systems, Bulletin 4021566.
2. Inspect aftertreatment DEF dosing system component electrical connections for any sign of corrosion.
3. Install aftertreatment DEF dosing system components, if previously removed.
4. Fill aftertreatment DEF tank with DEF (optional).
5. Activate DEF dosing system. Use INSITE™ electronic service tool to perform Diesel Exhaust Fluid Doser Pump Override Test to verify flow volume is within specification. See corresponding Service Manual. Reference Procedure 011-063 in Section 11.
6. Install aftertreatment DEF dosing valve. See corresponding Service Manual. Reference Procedure 011-059 in Section 11.

Note : For UL2 and UL2.2 aftertreatment DEF dosing systems, check valves in aftertreatment DEF dosing unit may stick when put back into service; use shop air to unstick check valves. See corresponding Service Manual. Reference Procedure 011-143 in Section 11.

Document History

Date	Details
2018-1-18	Module Created

Date	Details
2020-8-11	Update with optional and alternative steps to reflect variation in customer preferences to not remove the DEF system from the vehicle for long term storage.
2023-3-16	Added mention of UL2.2 throughout service bulletin.

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